

To relax the constraint $x_1 + x_2 = 2$, we first write it as $2 - x_1 - x_2 = 0$. Indeed, the quantity $2 - x_1 - x_2$ quantifies the amount of violation of the constraint. If it is zero, the constraint is not violated. If it is different from zero, it is. We introduce a penalty that is proportional to the violation of the constraint, with a proportionality factor λ . The new optimization problem is

$$\min_{x \in \mathbb{R}^2} x_1 + 4x_2 + \lambda(2 - x_1 - x_2)$$

subject to

$$\begin{aligned} x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

We examine the different cases corresponding to the different values of λ .

- If $\lambda = 0$, the problem is

$$\min_{x \in \mathbb{R}^2} x_1 + 4x_2$$

subject to

$$\begin{aligned} x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

Its optimal solution is $x^* = (0, 0)^T$ with a value of 0. This solution violates the constraint of the original problem, and the optimal value is lower.

- If $\lambda = 1$, the problem is

$$\min_{x \in \mathbb{R}^2} 2 + 3x_2$$

subject to

$$\begin{aligned} x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

Any feasible x such that $x_2 = 0$ is an optimal solution to the problem with a value of 2, which is the same optimal value as the original problem, with the constraint. Because of the penalty, there is no incentive to violate the constraint, as it does not allow to improve the solution of the original problem.

- If $\lambda = 2$, the problem is

$$\min_{x \in \mathbb{R}^2} 4 - x_1 + 2x_2$$

subject to

$$\begin{aligned} x_1 &\geq 0, \\ x_2 &\geq 0. \end{aligned}$$

The problem is not bounded, as the value of the objective function goes to $-\infty$ when x_1 goes to $+\infty$. The value of the penalty parameter is not appropriate, as it transforms a well-defined problem into an unbounded problem.